KETOROLAC

Max 5 days total

High GI/renal risk

CHEMICAL NAME (±)-5-benzoyl-2,3-dihydro-1H-pyrrolizine-1-carboxylic acid (as trometamol salt)

CLASS Non-selective NSAID (pyrrolizine carboxylic acid derivative)

TRADE NAMES Toradol · Ketorol · Acular (eye drops)

THERAPEUTIC USES / INDICATIONS

- Short-term management of moderate to severe acute pain (often post-operative).
- Renal colic.
- Eye drops: post-cataract inflammation, allergic conjunctivitis.

ADMINISTRATION

- **IV:** bolus over at least 15 seconds.
- **IM:** slowly and deeply into muscle.
- **PO:** only as continuation of IV/IM therapy.
- Total duration (all routes) must not exceed 5 days.

DOSAGES

- **Adult IV/IM:** 10–30 mg every 4–6 h; max 90 mg/day (60 mg/day in elderly, <50 kg or renal impairment).
- **PO:** 10 mg every 4–6 h; max 40 mg/day.
- **Children (IV, single dose):** 0.5 mg/kg (max 15–30 mg).

PHARMACOKINETICS

- **Onset:** IV/IM 10–30 min. **Peak:** 1–2 h. **Duration:** 4–6 h.
- Highly protein-bound; metabolised in liver.
- Excreted in urine. **t½:** 5–6 h (longer in elderly).

PHARMACODYNAMICS

- Potent inhibition of **COX-1 and COX-2.**
- Analgesic potency comparable to moderate opioid doses, without respiratory depression; modest anti-inflammatory effect.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, dyspepsia, abdominal pain.
- Drowsiness, headache, sweating.
- Injection-site pain; oedema.

ADVERSE EFFECTS (SERIOUS)

- GI bleeding and perforation (high risk).
- Acute kidney injury.
- Post-operative bleeding (antiplatelet effect).
- Anaphylaxis; bronchospasm.

CONTRAINDICATIONS & CAUTIONS

- Active or past peptic ulcer, GI bleeding.
- Moderate–severe renal impairment; hypovolaemia.
- Bleeding disorders, cerebrovascular bleeding, high-bleeding-risk surgery.
- Labour and delivery, breastfeeding; use with other NSAIDs or aspirin.

KEY INTERACTIONS

- **Other NSAIDs/aspirin, anticoagulants:** contraindicated combinations.
- **Probenecid, pentoxifylline:** contraindicated.
- **Lithium, methotrexate:** ↑ levels.

NURSING MANAGEMENT

- Confirm hydration status and renal function before giving.
- Record start date — stop after 5 days.
- Monitor urine output, BP, wound bleeding and stools.
- Assess pain 30 min after dose.
- Watch for allergy, especially in asthmatic patients.

PATIENT EDUCATION

- This drug is for short-term use only.
- Report reduced urine, black stools or bleeding.
- Do not take other painkillers containing NSAIDs.
- Drink adequate fluids.

KEY POINTS

- Strongest analgesic NSAID; opioid-sparing.
- High risk of GI bleeding and kidney injury.
- Max 5 days total, all routes.
- Reduce dose in elderly and <50 kg.